

Spatial and Temporal Analysis: Crime Patterns in Washington D.C, A Data Driven Approach to Urban Safety

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Abstract

Crime has always been a large issue in the United States and higher rates of incidence tend to occur within a concentrated area of a city. This study explores crime patterns and weapon usage across the Washington D.C. area in 2024 alone using incidence reports from local police departments. Geo-Mapping was utilized to visualize incidence levels per wards and certain areas. This research contributes to the previous workings of crime occurrence in the Washington D.C. area and around the United States of America in a more focused level. The key finds from this paper has revealed that incidence is more prominent during the evenings, however the difference between evening and night is small based on the large amounts of reports and violent crimes may be disproportionately concentrated in certain areas and wards. The insights derived from this paper can allow for data-driven decisions in policing strategies and urban development.

I. INTRODUCTION

Crime remains a hot topic and issue in major cities including areas within Washington D.C. especially as the nation's capital. Though crime appears to still be prominent, there seems to be a drop in crime yearly according to the data the Metropolitan Police Washington DC

updated as part as an initiative for transparency for the public. Violent crime seems to have decreased overall by around 35% with the highest drop in robbery. [1] Combatting crime will always be a challenge that law enforcement must face, however knowing hotspots of crime within the city allows for increased efficiency in police stationing and enforcement. There was once a time when Washington DC was known as the "murder capital of the US" in the 1900s when murder has risen to an alarming degree due to the uptick of gun violence and drug abuse [2], crime is no longer as bad, however there is still a large number of crimes that occurs within certain districts of DC.

Controlling crime by larger police presence is a well-known strategy employed by the police especially in areas with more crime than others. There are certain areas such as Northwest D.C. where there are lots of exposure to homicides within a half of mile radius while areas such as western D.C. has little to no exposure to homicides [3], thus there is historical evidence that disproportionate amounts of violent crimes occur in different neighborhoods. This study aims to provide insight on certain areas of Washington D.C. to explore trends, correlations and distributions of reported incidence. This paper aims to answer important questions such as what offense was committed the most and how was that crime committed, do certain areas of Washington D.C. have a higher concentration of crimes compared

to other areas, and are crimes more likely to be reported during the mornings or nights? The paper aims to answer the problem statement:

What reported crimes are the most common and what district/wards does it occur the most?

The paper is structured as follows: Section II: Literature Review of related works done by other researchers. Section III, a description of the dataset that is being used and where it was sourced. Section IV, data framework for analysis of the dataset. Section V, discussions and in-depth analysis of the dataset. Section VI, is the discussion. Section VII, is the conclusion, and section VIII will be the references.

II. LITERATURE REVIEW

There has been many previous research on analyzing crime statistics throughout the United States and the correlation of areas with high weapon and alcohol ownership towards crime rates per area. From one research that was conducted by Franklin, LaVeist, Webster, and Pan, it was deduced that if an area had a high prevalence of weapons as well as illicit drugs, a correlation can be made towards assault with a deadly weapon, a violent crime. [4] With this revelation, looking for certain areas in Washington D.C. that has the highest rate of weapon ownership or areas that are classified as “dangerous” can indicate the type of crime that is most common in the district. The researchers in this case used census data to determine the crime levels, however it does not delve deep into the causes and it uses census data as a proxy to community neighborhoods which may not give an accurate result. Another research that was conducted by Irvin-Erickson and Vigne, their findings was that there are higher crime rates (56%) within modes of transportation at off-peak hours, this means that there are more violent crimes that occur when there are less people around as well as less security. [5] This is a key indicator that crimes occur within the night or very early morning which can indicate a trend to occur within the overall crime rates. This research conducted provides evidence that the times can have a major effect on when

crimes happen. There was a correlation that was made between time of day and crime rate within public transportation modes within Washington D.C. This study although it provides evidence of correlation, it seems to have surveyed from a limited region within the District of Columbia which can serve to be too small of a sample size as it is too concentrated in one area. A study that was conducted by Ee and Zhang focused on the homelessness problems in the United States. The researchers used OLS regression to both derive predictions and display the current trends of homelessness and violent crimes. They found that a larger volume of homelessness within an area does in fact have a positive relationship to the number of crimes that are committed. [6] Though this study was not directly related towards the target area (District of Columbia) it does provide necessary insight of how a larger population of homeless within an area can be means of higher violent crime rates, and there are most certainly areas within the District of Columbia where the issue of homelessness is a large problem. The research paper gives us insights on trends in areas around the United States, not only just the District of Columbia. From the use of Ordinary Least Squares Regression, a model must be fitted, and fitting the models can result in multicollinearity which this study may have occur. Often any social disorganization variables such as poverty, unemployment, education level can have a high correlation with one another thus, distorting regression coefficients, this can make it difficult to interpret the effects each variable truly has towards the target variable which in this case is number of violent crimes. These sources do hold relevance to the research questions that is being asked, such as the prevalence of crime during day or night, which areas tend to have a higher concentration of crime, and areas where homelessness is prevalent and the effect on violent crime. These variables presented by the research done can provide major detail on which areas may be needed for higher levels of police presence and knowledge on the types of crimes typically committed.

III. DATASET

The dataset used for this research was sourced from data.gov published by the District of Columbia Metropolitan Police Department. [7] The data contains information of crime records in 2024 and some records of the beginning of 2025. This dataset contains 26 columns and 29294 rows of data on crime reports in the District of Columbia. The notable columns included are the method of crimes committed, offense, and ward. Table 1 depicts the NOIR datatypes (ordinal, interval, ratio, nominal) as well as the attributes (column names) of the dataset, starting from geo-data to categorical data such as offense committed, district, method, etc.

Figure 1

Table 1: Crime Dataset Attribute Types

Attribute	Data Type	Attribute	Data Type
X	Ratio	XBLOCK	Ratio
Y	Ratio	YBLOCK	Ratio
CCN	Nominal	WARD	Ordinal
REPORT_DAT	Interval	ANC	Nominal
SHIFT	Ordinal	DISTRICT	Ordinal
METHOD	Nominal	PSA	Ordinal
OFFENSE	Nominal	NEIGHBORHOOD.CLUSTER	Nominal
BLOCK	Nominal	BLOCK.GROUP	Nominal
CENSUS.TRACT	Nominal	VOTING.PRECINCT	Ordinal
LATITUDE	Ratio	LONGITUDE	Ratio
BID	Nominal	OBJECTID	Nominal
START_DATE	Interval	END_DATE	Interval
OCTO_RECORD_ID	Nominal		

IV. DATA FRAMEWORK

This research paper focuses on the usage of 3 technologies to perform statistical analytics and visualization as well as the use of libraries such as seaborn and ggplot2. The main technologies focused on is Python, R, and Amazon Athena SQL. Python and SQL are used to delve into the data and perform data exploration and analytics. While R is used mainly for visualization purposes and overall readability of the data. Data exploration was done for the shape, datatypes, and overall data to find any correlation as well as grouping the data to find out which crime is committed the most, where it is committed, the shift it has occurred in, etc. Visualization using R was done for mapping the Washington D.C. area based on crime density, overview of the times that reports are made (day, evening, midnight), offense types per ward using a grouped bar chart for better clarity and visual of the top 3 offenses, etc. The

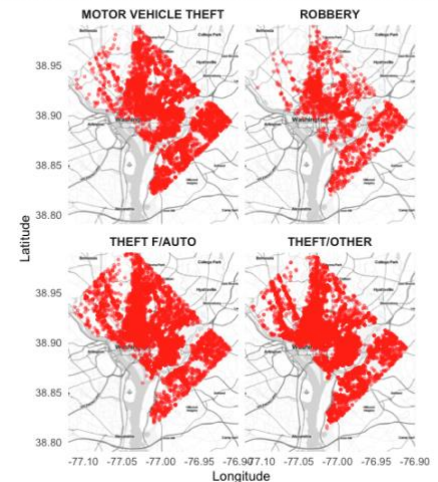
purpose of visualizing the data as a map gives easily accessible insights of most common crime locations, and the bar charts that were created allow for easy comparisons of data points regarding wards and offense types. SQL (Amazon Athena) was used for surface level analysis and answering the research questions of where/when crimes usually occur, weapons that are most prevalent in reports, etc. Python was used to create visualizations supporting R visualizations, surface analysis, and statistical analysis. Python was also used to run quick summaries on the dataset. Though visualizations are a main point to make raw data easily digestible, some summaries will not benefit from visualizations and will be presented in tabular format.

V. INCIDENCE ANALYSIS:

I. Varying crime per location

Figure 2

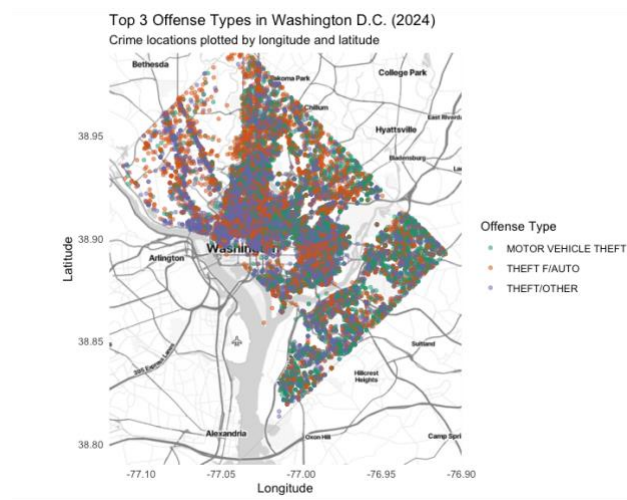
Top 4 Crime Types in Washington D.C. (Individual Incidents)



The top 4 crimes that occur within Washington D.C. are motor vehicle theft, robbery, theft f/Auto and other types of theft. Certain parts of D.C. have varying levels of reports such as the area around Georgetown and Palisades (near American University) has lower degrees of general robbery and motor vehicle theft, however theft F/Auto (Theft from Auto, stealing items from parked vehicle) is more common around that general vicinity. A likely

indicator may be that college campuses tend to be safer with higher police presence and school security, thus leading to the decrease in criminal activity, however based on both figure 2 and 3, crimes do occur in every area, some less than others.

Figure 3



Green represents motor vehicle theft, red represents theft F/auto, and Blue is other theft.

Though there are crimes that occur in almost every area of D.C. overall, there are still some spots that have not had any or minimal numbers of crime reported, which may be due to the security of the area or if the area is well funded.

Figure 4



Motor vehicle, theft F/auto, theft/other respectively.

There are areas where crime is heavily concentrated as shown on figure 4. All crimes ranging from motor vehicle theft, other types of theft, which may indicate a need for higher police presence and shows that more criminal

activity happens in certain areas rather than being spread out.

II. Common crimes per time and place

Crimes occur throughout the city, however theft categorized as “other” are most common throughout every ward and district, with heavy prominence as alluded from figure 5 grouped bar plot. Ward 2 has an unusual amount of crime far above the average for crimes of “other” theft, which may be an indicator of crime hotspot or an area below the lines of poverty. Though ward 1 also has a large number of concentrated crimes regarding theft, ward 2 has a far larger amount potentially indicating large crime activity.

Figure 5

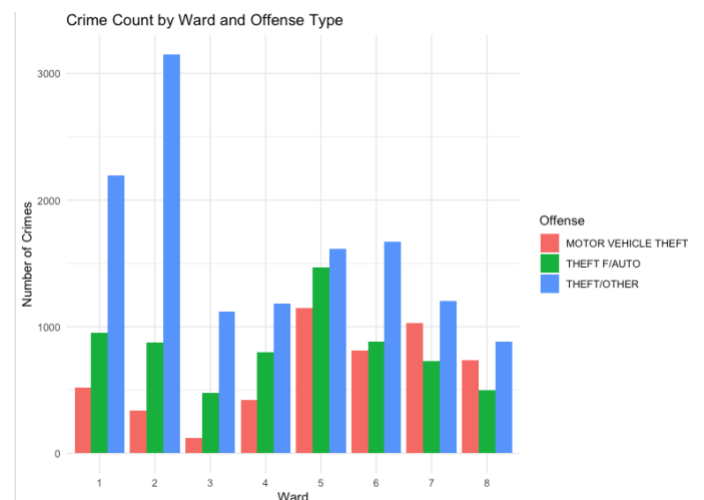
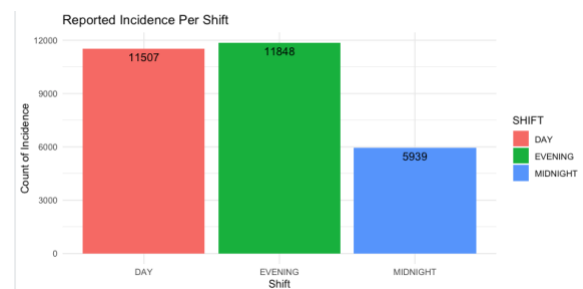


Figure 6



Times where most crimes are reported have much spread, however there are points where the number of reports peak. Based on

hourly reports, there are much more reports at around the 17th hour (5 PM) point with 1898 crimes reported, while the lowest occurs at hour 10 (10 AM) at 535. In 2024, there appears to be more crimes occurring during peak hours, rather than overnight or during the mornings as represented by figure 7. From the 10th hour (the lowest point), the number of reported incidences increases about evenly up until the peak at hour 17. Other than the gradual increase in reports, the incidents are more spread out.

Figure 7



Crimes tend to occur all throughout the day, however the times that are reported such as day or evening, seem to have a common method, that is “OTHERS” which are categories of crime that do not fit into the traditional categories of murder, theft, assault, homicide, etc. Some examples of “other” classifications may be trespassing, vandalism, noise complaints, etc. “Other” forms of methods occur throughout the day the most at 11029 reports and evenings at around 10955 reports, a minuscule difference. On figure 8, aside from other methods of crime being the most popular, gun violence cases are prevalent as well. This can be due to the wide availability of firearms in the United States thus causing an increase in firearm related cases throughout the United States, not just Washington D.C.

Figure 8

method	shift	method_count
OTHERS	DAY	11029
OTHERS	EVENING	10955
OTHERS	MIDNIGHT	4718
GUN	MIDNIGHT	1054
GUN	EVENING	741
GUN	DAY	380
KNIFE	MIDNIGHT	167
KNIFE	EVENING	152
KNIFE	DAY	98

Overall, crimes are reported the most during the day-midday times by a large margin, however the number of reports for daytime and evenings for the most popular method of crime are about equal, though day shifts have slightly more reports. This can be attributed to the amount of variety that comes with different types of reports that does not have to do with violent crimes (e.g. assault with a deadly weapon, gun violence, knife violence, etc.).

III. Most Committed Crimes

There are certain crimes overall that have more incident reports than others. The category of “theft/other” have the most reports at 13014. In figure 9, theft occurs the most compared to other offenses, however other crimes such as robbery and assault with a deadly weapon are not as commonly reported. This can be attributed by the fact that theft occurs when there are large amounts of tourism and high population density. However, the underreporting of other crimes can also play a part in the data not being up to date.

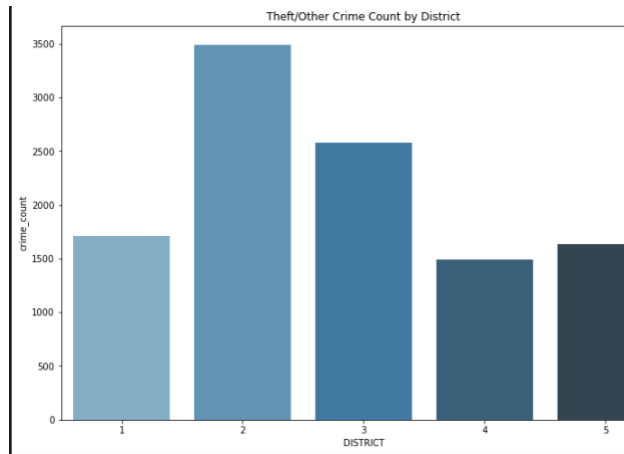
Figure 9

offense	crime_count
THEFT/OTHER	13014
THEFT F/AUTO	6680
MOTOR VEHICLE THEFT	5126
ROBBERY	2109
ASSAULT W/DANGEROUS WEAPON	1027

Washington D.C. district 2 has the most crimes filtered by the most common crime of Theft/other by over 1000 reports. This can be a clear indicator that there may be high levels of poverty or high reporting patterns by the civilians that live in that area. District 3 has more crime above the average for theft/other crimes, compared to districts 1,4, and 5, however the overwhelming number of reports in

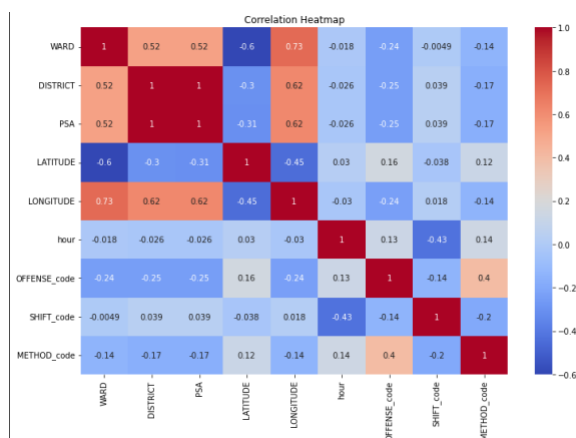
district 2 can be cause for concern about the overall crime rate in that area.

Figure 10



There is a medium correlation between the method of the crime and the type of offense that is done. However, there does not seem to be a correlation between any of the variables otherwise that can potentially be used to predict the outcome of the crimes committed by area or the type. Though there are correlations between values such as ward and district, PSA, etc. it does not provide evidence of the type of crime and the area it is occurring. It is known that these location-based attributes are all correlated with one another.

Figure 11



VI. DISCUSSION

The analysis of crime incidence in Washington D.C. for 2024 provided valuable insights on what type of offenses are the most common and what can be potentially done about it. Though there are various crimes that have been reported and recorded, one stood out the most, that is offenses labeled “Theft/other”, which are shoplifting, pickpocketing, package thievery, etc. To answer the questions of “do certain areas of Washington D.C. have a higher concentration of crimes compared to other areas?” There is evidence that supports this case, ward 2, district 2, appears to have the highest amounts of cases of “theft/other” (the most popular report) reports compared to other wards and districts which can be a clear indicator of socioeconomic or policy issues that may need to be addressed.

For the second research question, “what offense was committed the most and how was that crime committed” The most common offense was theft in the other category by a large margin, this can be attributed to the variety of reports that may fall into this category that do not fall within other categories such as homicide, auto theft, or robbery. These crimes committed use a variety of weapons such as guns, knives, however most of the methods that was used for these categories is also labeled as “other” which can mean fists, poles, or other weapons that are not guns or knives.

For the last research question “are crimes more likely to be reported during the mornings or nights?”, it appears that crimes are more likely to be reported around mornings to mid-day and not off-peak hours. From the line-graph it seems that the crime reports are around the 17th hour (5 PM), which can be an indicator that security or police presence may be needed around these times. The methods that were used during the mornings are “others” as well as the evenings, which can show that traditional weapons such as guns and knives are not as prominent as one may think. It also appears that time does not really play too much of a role in crimes reported as both counts are about the same with one another, however daytime reports are slightly more common.

VII. CONCLUSION

The study was done with a publicly available dataset sourced from data.gov [7] and published by the District of Columbia Metropolitan Police Department. Multiple technologies were used (Python, R, AWS Athena SQL) to visualize, summarize, and derive insights from the raw data. Crime is prominent in every country/state, Washington D.C. is no exception, therefore finding insights and trends of the crimes that are committed can help police/relevant authorities to create policies and police placement in order to maintain order as well as promote urban safety. Using data to determine where and how offenses are committed and the prominence of such crimes can allow authorities to create a countermeasure against crimes and to improve efficiency to how police are prepared and the response time may also be greatly improved. As a further study, there can be more in-depth analysis on past, present, and future years such as comparisons of historical crime data in Washington D.C. and perhaps future predicted crimes. A limitation of this study is the lack of predicted analysis done as well as the limited data that was used. Having a larger dataset that contains more predictive variables will allow for more in-depth analysis and more accurate results that can be applied to further improvements to police activity and monitoring.

VIII. REFERENCES

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